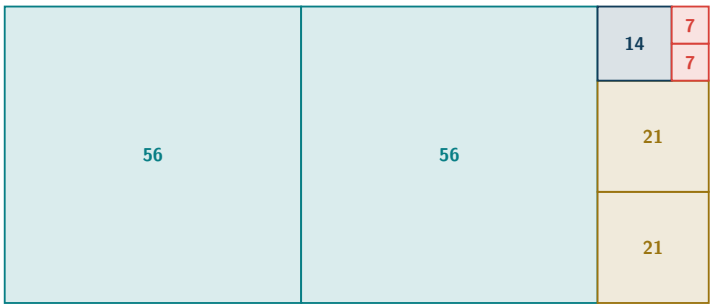
NUMBER THEORY LADDER 1 OF 9

Euclid's algorithm and Bézout's identity

Repeated remainders find the HCF, and running them backwards writes it as $ax + by$.

1 / THE RULE AND THE PICTURE

The HCF of two numbers is unchanged when the larger is replaced by its remainder



a 133×56 rectangle cut into the largest squares, and the smallest square is $7 = \mathrm{HCF}$

Any number dividing a and b also divides $a - qb$, and conversely, so the common divisors of a, b and of b, r coincide:

$$\boxed{\mathrm{HCF}(a, b) = \mathrm{HCF}(b,\ a \bmod b)}$$

Cutting off squares removes qb from a at each step.

2 / THE ALGORITHM / 133 AND 56

Divide, keep the remainder, and stop when it reaches zero

a	b	q	r	division
133	56	2	21	$133 = 2 \cdot 56 + 21$
56	21	2	14	$56 = 2 \cdot 21 + 14$
21	14	1	7	$21 = 1 \cdot 14 + 7$
14	7	2	0	$14 = 2 \cdot 7 + 0$

The last non-zero remainder, 7, is $\mathrm{HCF}(133, 56)$, and since $133 \cdot 56/7 = 1064$ the lowest common multiple follows from $\mathrm{HCF} \times \mathrm{LCM} = ab$.

3 / BÉZOUT'S IDENTITY / BACK-SUBSTITUTION

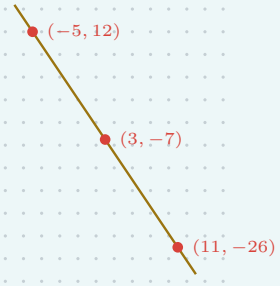
Working backwards writes the HCF as $ax + by$, and its solutions form a line of lattice points

Using each remainder in turn,

$$\begin{aligned} 7 &= 21 - 14 = 21 - (56 - 2 \cdot 21) = 3 \cdot 21 - 56 \\ &= 3(133 - 2 \cdot 56) - 56 = 3 \cdot 133 - 7 \cdot 56, \end{aligned}$$

so $x = 3$ and $y = -7$ solve $133x + 56y = 7$, and adding $(56/7,\ -133/7) = (8, -19)$ gives all solutions:

$$\boxed{(x, y) = (3 + 8t,\ -7 - 19t), \quad t \in \mathbb{Z}}$$



4 / WHAT IT GIVES / SOLVABILITY, INVERSES AND THE LCM

$ax + by = c$ has integer solutions exactly when $\mathrm{HCF}(a, b)$ divides c

So $133x + 56y = c$ is solvable when $7 \mid c$ and never otherwise.
If $\mathrm{HCF}(a, n) = 1$ then $ax + ny = 1$ and x is the inverse of a modulo n: from $10 \cdot 4 - 13 \cdot 3 = 1$ the inverse of 10 modulo 13 is 4, the multiplier behind the divisibility test for 13, and $10 \cdot 5 - 7 \cdot 7 = 1$ gives the multiplier 5 for 7.
The inverse of 56 modulo 133 does not exist, because the HCF is 7.

5 / A SECOND EXAMPLE / 1071 AND 462

The same steps give $\mathrm{HCF} = 21$ and the identity $21 = 7 \cdot 462 - 3 \cdot 1071$

$$\begin{aligned} 1071 &= 2 \cdot 462 + 147 \\ 462 &= 3 \cdot 147 + 21 \\ 147 &= 7 \cdot 21 + 0 \end{aligned}$$

Back-substituting, $21 = 462 - 3 \cdot 147 = 462 - 3(1071 - 2 \cdot 462) = 7 \cdot 462 - 3 \cdot 1071$, and a check gives $3234 - 3213 = 21$. Every linear combination $1071x + 462y$ is a multiple of 21, and every multiple of 21 occurs.

ASIDE / FAST IN PRACTICE

The remainders at least halve every two steps, so the algorithm takes at most about $\log_2$ of the number of steps in digits.

LINKS TO OTHER SHEETS

The ladder continues with *Prime factorisation, divisors and squares*; the divisibility multipliers appear in *Primes, the sieve and divisibility tests to 13*.

Convention. HCF is the highest common factor (GCD); $a \bmod b$ is the remainder; t ranges over the integers.

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